Arimitsu Horikawa-Strakovsky
859-539-3983 | arimitsu0803@gmail.com | arimitsu06.github.io

Education

California Institute of Technology **Expected** June 2029
Bachelor of Science in Computer Science, Minor in Biology, **GPA**: 4.0 *Pasadena, CA*
- **Coursework:** Data Structures & Algorithms, Operating Systems & Computer Architecture, Machine Learning & Data Mining, Linear Algebra, Differential Equations, Discrete Math, Organic Chemistry, Classical Mechanics, Electromagnetism & Special Relativity

Experience

California Institute of Technology — Anima AI+Science Lab Nov. 2025 – Present
SURF Fellow Summer '26, AI/ML Researcher *Pasadena, CA*
- Co-leading development of **AI/deep learning** algorithms for hybrid density functional theory (**quantum chemistry**), matching PBE0 reaction barrier heights to **<1 kcal/mol** while reducing computational complexity from $O(N^4)$ to $O(N^3)$ & using semi-local orbitals
- Built **benchmarking & evaluation pipelines** comparing orbital-learning models to **Meta FAIR's UMA**, showing flat error vs. UMA's exponential error growth on increasingly positively charged molecules, resulting in **5th authorship** on a *Nature Comp. Sci.* paper
- Own the molecule-focused branch end to end (architecture modifications, physics-informed losses, dataset curation, unit tests, ablations, debugging), contributing ∼**5,000 lines** to a ∼40,000-line **Python & PyTorch** codebase shared by a 5-person team, delegating well-scoped features & scripts to AI coding agents/Claude Code with unit tests for verification
- Develop & maintain data-generation, **distributed AI training**, & evaluation pipelines (**SQL, Slurm, Linux, Shell, PySCF**) running on **NVIDIA H200/B200 GPUs** across two **HPC** clusters, with versioned code state (**WandB**) for reproducible runs
- Sideloaded analytic density gradients into the input pipeline & rerouted backpropagation through computed density gradients, cutting the physics-constraint error floor **1.4–2**× vs. 6th-order finite differences;
- **Profiled** training with **PyTorch Profiler** & found redundant autograd recomputation across physics losses as the main bottleneck; built a shared autograd cache & unified floating-point precision without destabilizing physics losses, speeding up training **1.8**×
- Traced unphysical exponential asymptotics to softplus output activation, evaluating fixing asymptotics with identity activation and Fermi-Amaldi potentials

SPARC Labs June 2026 – Present
Research Director *Pasadena, CA*
- **Direct research** at Caltech's first student-led **AI** research lab, supervising projects & selecting proposals from **over 50 members**
- Supervised & mentored **3 research projects** from proposal to acceptance at **6 NeurIPS 2026 workshops**; currently supervising 4
- Projects range LLMs, Vision-Language Model (VLM) alignment (colab with **Amazon Science**), AI4Math, robotics pathfinding, & more

University of Kentucky ICU Recovery Clinic Jan. 2025 – Jan. 2026
Research Analyst I (Technical Lead) *Lexington, KY (Remote from Sep. 2025)*
- Developed **computer vision models** that automatically measure biceps & quadriceps **ultrasounds** from a hand-held **sensor**, cutting analysis time by **99.3%**; co-developed with a clinical partner in New Zealand
- Built & shipped a **Napari/Qt desktop app** (∼2,700 lines of **Python**) serving **5-model PyTorch ensembles** with **TorchScript** inference, containerized with **Docker**; adopted by clinicians at **3 universities**, who now run AI analysis without engineering support
- Designed a **human-in-the-loop** workflow: AI outlines that users can edit, updating measurements automatically; **EasyOCR**-based scale calibration; & **batch inference** with automatic failure flagging & resumable sessions

University of Kentucky Institute for Biomedical Informatics Nov. 2023 – Aug. 2025
AI/ML Researcher *Lexington, KY*
- **Solo developer** of MyoVision-US (∼3,500 lines of **Python/PyTorch**), an **AI semantic segmentation** tool for muscle ultrasound analysis that reaches **ICC 0.85-0.99** against expert physiotherapists & cuts muscle ultrasound analysis time by **99.8%** (24 h → 4 min for 180 images), resulting in a **co-first authored** paper in *Scientific Reports*; built & hosted a public demo on **Hugging Face** spaces
- Curated a **multi-site training set** (4 ultrasound devices across the US, Australia, & Chile) & trained models end to end, **coordinating labeling & failure-mode analysis** with clinicians across multiple model iterations
- Ran **ablation studies** across 4 architectures (**DeepLabV3**, **U-Net**, **SegFormer**), 2 augmentation schemes, & 5 loss functions using **PyTorch** & **Albumentations**, reaching **0.90/0.95 Dice** (quadriceps complex/tibialis anterior)
- Deployed MyoVision-US to **6 clinical research projects** at universities in the US & Australia, leading to **authorship on 3 additional published papers**

Selected Publications — ∗ denotes co-first authorship with equal contribution

- Kang, B. S., . . . , **Horikawa-Strakovsky, A.**, . . . , & Anandkumar, A. OrbitAll: a unified quantum mechanical representation deep learning framework for all molecular systems. [Accepted to *Nature Computational Science*]
- *Rivera, Z. C., *González-Seguel, F., ***Horikawa-Strakovsky, A.**, . . . , & Wen, Y. Development of an artificial intelligence powered software for automated analysis of skeletal muscle ultrasonography. *Scientific Reports*
- White, M. S., **Horikawa-Strakovsky, A.**, . . . , & Wen Y. Open-Source AI for Vastus Lateralis and Adipose Tissue Segmentation to Assess Muscle Size and Quality. *Ultrasound in Medicine & Biology*

Projects

Continual Learning AI Research | *Python, Pytorch, PyCIL* Nov. 2024 – May 2025
- Developed an improved self-supervised label augmentation (SSLA) method for exemplar-free class-incremental learning, outperforming the previous state-of-the-art, AdaGauss (**NeurIPS '24**) by **+2.5 pts on CIFAR-100** & **+2.7 pts on Tiny-ImageNet** on last task accuracy; earned **4th Grand Award** in Robotics & Intelligent Machines at **Regeneron International Science & Engineering Fair**
- Limited **knowledge distillation** to non-augmented logits & applied SSLA only on the first task, gaining **+7.4 pts** while cutting training time **2.4**×; designed an **FFT**-based frequency-domain augmentation that matched rotation-based SSLA within 0.8 pts
- Ran **70 experiments** across 3 datasets & 5 random seeds using **PyTorch** & **PyCIL**, with ablations & training logs released **open-source**

2x Kaggle Expert | *Python, PyTorch, TensorFlow, NumPy, SciPy, Pandas* May 2021 – May 2023
- Placed **17th of 801 teams (top 2%; Silver Medal)** in **BirdCLEF 2022** (bird-call audio classification) on a 2-person team; trained a 25-model **CNN** ensemble (5 backbones × 5 folds, incl. NFNet, EfficientNetV2) on **mel-spectrograms** using 2× NVIDIA A40 GPUs
- Published Jupyter notebooks ranging **exploratory data analysis, image classification, object detection, clustering, and instance segmentation** with **4,500+ forks** and **2,000+ upvotes**, reaching a **top 0.9%** peak notebooks ranking and a **top 0.5%** peak competitions ranking

Technical Skills

Languages: Python, Java, C, SQL, ARM Assembly, JavaScript, HTML/CSS
ML & Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, Agent ReAct Loop, OpenCV, PySCF, tblite, MMDetection, Detectron2
Developer Tools: Linux, Git, Docker, Slurm/HPC clusters, Shell, SSH, Multiprocessing, AI coding agents, Claude Code, GitHub
Concepts: Deep Learning, Machine Learning, Custom Agent Harnesses, Exploratory Data Analysis, Dataset Curation, Data Pipelines, Human-In-The-Loop Design, CI/CD Development, Computer Architecture, Operating Systems, Multiprocessing, CPU Architecture, GPU Acceleration, GPU Computing, Compiler Programming, Version Control